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Trajectory Tracking Controller Design for Caterpillar Vehicles Using a Model Reference Adaptive Controller

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Abstract: This paper proposes a trajectory tracking controller for Caterpillar Vehicles (CVs) with uncertain parameters that combines a kinematic controller based on a backstepping method and a dynamic controller based on a model reference adaptive controller (MRAC). To do this task, the followings are done. Firstly, a system configuration of the CV is described. Secondly, kinematic modeling and dynamic modeling of the CV with some uncertain parameters are presented. Thirdly, a kinematic controller is designed based on kinematic modeling such that a local posture tracking error vector converges to zero. Fourthly, a dynamic controller based on MRAC is designed based on dynamic modeling such that CV's output velocity vector converge to a virtual control input vector. Thus, a trajectory tracking controller using backstepping-based MRAC is utilized to estimate these uncertain parameters. By choosing a suitable Lyapunov function candidate, system stability is guaranteed, and a control law and update laws are obtained. Finally, the simulation and experimental results are presented to verify the effectiveness of the proposed MRAC controller.

Keywords: CV, MRAC, trajectory tracking controller.

1. INTRODUCTION

Recently, autonomous robots have been used instead of human beings. The mobile robots are able to accomplish various tasks in dangerous places where workers cannot enter. To solve the above tasks, mobile robots as a Caterpillar Vehicles (CVs) were chosen. Especially, unlike the mobile robot with a conventional wheel, CV can work in rough terrain environments. The CV was considered as the most important type of mobile robots for applications in all-terrain environments. It has two tracked wheels on the left and right sides of the CV's body. The tracked wheel is a special type of wheels for all-terrain conditions. A large contact area of tracks with the ground provides superior advantages for the CV such as better mobility in unstructured environments. The CV has good performance in moving over obstacles and adapting to rough terrain environments such as running on the grass, ditch, sands, and stairs quickly. Therefore, the application of the CV systems has been increasing in different areas during recent years. Trajectory tracking control problem is one of the key techniques in mobile robot research. Thus, several types of methods were used for tracking control of the mobile robot in [1-6]. M. V. Gomes et al. [1] proposed a PID controller for trajectory tracking of two-wheeled mobile robots. S. C. Yuan et al. [2] proposed a fuzzy control method for a two-wheeled mobile robot. G. Oriolo et al. [3] proposed a dynamic feedback linearization control for a two-wheeled mobile robot (WMR). N. Hung et al. [4] proposed a sliding mode controller design for two-wheeled mobile robots. P. S. Paratama et al. [5] presented a backstepping controller for a two-wheeled automated guided vehicle in a partially known environment. These previous papers have been applied for mobile robots with two conventional wheels but there were not applied for a mobile robot with a caterpillar type of wheels yet. The tracking controller design problem for a mobile robot with a caterpillar type of wheels is a challenging mission due to the system modeling of the Caterpillar Vehicle with some uncertain parameters. In practical applications, in order to design a trajectory tracking controller for mobile robots, a model with all known parameters is needed. However, some physical parameters are difficult to be computed or identified for constructing a system model. Although some identification techniques can estimate unknown parameters of a system, they require a high cost of typical hardware and software. To solve this problem, a control method called as model reference adaptive control (MRAC) was developed for various unknown systems. E. Canigur et al. [7] presented MRAC for trajectory tracking of a wheeled mobile robot (WMR) with uncertain parameters in a dynamic model. R. Prakash et al. [8] presented modeling and simulation for a fuzzy logic controller using MRAC. However, they only presented simulation results. Moreover, the previous researches for the CV in [10-13] considered only the kinematic model which ignored the mechanical system dynamics and considered only the desired velocity vector as the system input, and so couldn't provide good trajectory tracking performance.

Therefore, this paper proposes a trajectory tracking for the Caterpillar Vehicles using the backstepping-based MRAC to estimate uncertain parameters of the system modeling of the Caterpillar

Vehicle. To verify the effectiveness of the proposed methods, the simulation and experimental results are shown.

2. SYSTEM MODELING

In this paper, the Caterpillar Vehicle uses a differential drive wheeled configuration. The schematic modeling of the system is shown in Fig. 1. In Fig. 1, *OXY* is a global coordinate frame. *Cxy* is the moving coordinate frame attached on the Caterpillar Vehicle (CV) platform. This CV has two driving standard wheels *W1* and *W2* with radius *r*, and *W3* and *W4* are passive wheels. (X_A, Y_A) is a CV position vector in global coordinate *OXY* with orientation angle θ_A. The CV moves with linear velocity V_A and angular velocity ω_A. v_R and v_L are the right and left wheel linear velocities, respectively. $\dot{\phi}_r$ and $\dot{\phi}_l$ are the right and left wheel angular velocities, respectively. *b* is a distance between wheels and *Cx* axis of the CV. *D* is a mass center of the CV. *d* is a distance between *C* and *D*.

The kinematic equation of the CV as shown in Fig. 1 can be expressed as follows:

$$\dot{\boldsymbol{x}}_v = \begin{bmatrix} \dot{X}_A \\ \dot{Y}_A \\ \dot{\theta}_A \end{bmatrix} = \begin{bmatrix} \cos\theta_A & 0 \\ \sin\theta_A & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_A \\ \omega_A \end{bmatrix} \Leftrightarrow \dot{\boldsymbol{x}}_v = \boldsymbol{H}(\boldsymbol{x}_v)\boldsymbol{\eta} \quad (1)$$

where $\dot{\boldsymbol{x}}_v$ is a posture vector of the CV in global coordinate *OXY*, $\boldsymbol{H}(\boldsymbol{x}_v)$ is a full rank matrix, and $\boldsymbol{\eta}$ is a velocity vector of the CV.

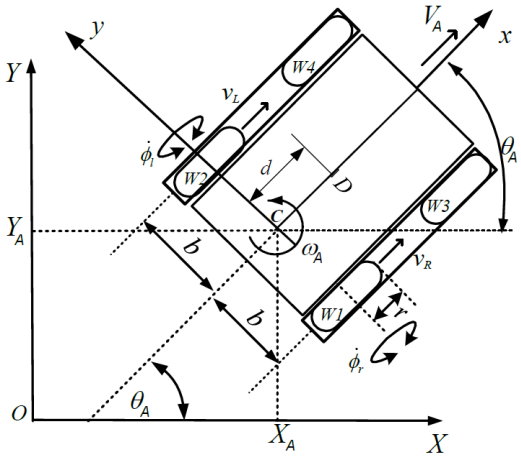


Fig. 1 Schematic diagram of the CV

From Eq. (1), the relation between a reference velocity vector in the global coordinate frame *OXY* and the reference velocity vector of the CV as follows:

$$\dot{\boldsymbol{x}}_r = \begin{bmatrix} \dot{X}_r \\ \dot{Y}_r \\ \dot{\theta}_r \end{bmatrix} = \begin{bmatrix} \cos\theta_r & 0 \\ \sin\theta_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_r \\ \omega_r \end{bmatrix} \Leftrightarrow \dot{\boldsymbol{x}}_r = \boldsymbol{H}(\boldsymbol{x}_r)\boldsymbol{\eta}_r \quad (2)$$

Relation of the linear velocity V_A and angular velocity ω_A with the right and left wheel angular velocities ($\dot{\phi}_r, \dot{\phi}_l$) can be expressed as:

$$\begin{bmatrix} V_A \\ \omega_A \end{bmatrix} = \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ \frac{r}{2b} & \frac{-r}{2b} \end{bmatrix} \begin{bmatrix} \dot{\phi}_r \\ \dot{\phi}_l \end{bmatrix} \Leftrightarrow \boldsymbol{\eta} = \boldsymbol{T}\dot{\boldsymbol{\Phi}} \ , \quad (3)$$

$$\Leftrightarrow \begin{bmatrix} \dot{\phi}_r \\ \dot{\phi}_l \end{bmatrix} = \begin{bmatrix} \frac{1}{r} & \frac{b}{r} \\ \frac{1}{r} & \frac{-b}{r} \end{bmatrix} \begin{bmatrix} V_A \\ \omega_A \end{bmatrix} \Leftrightarrow \dot{\boldsymbol{\Phi}} = \boldsymbol{T}^{-1}\boldsymbol{\eta} \ . \quad (4)$$

Dynamic equation of the CV can be described as follows [3]:

$$\bar{\boldsymbol{M}}\boldsymbol{T}^{-1}\dot{\boldsymbol{\eta}} + \bar{\boldsymbol{V}}\boldsymbol{T}^{-1}\boldsymbol{\eta} = \boldsymbol{\tau} \ , \quad (5)$$

$$\bar{\boldsymbol{M}} = \begin{bmatrix} \frac{r^2}{4b^2}(m_t b^2 + I) + I_w & \frac{r^2}{4b^2}(m_t b^2 - I) \\ \frac{r^2}{4b^2}(m_t b^2 - I) & \frac{r^2}{4b^2}(m_t b^2 + I) + I_w \end{bmatrix},$$

$$\bar{\boldsymbol{V}} = \begin{bmatrix} 0 & \frac{r^2}{2b} m_c d\dot{\theta}_A \\ -\frac{r^2}{2b} m_c d\dot{\theta}_A & 0 \end{bmatrix}, \boldsymbol{I} = m_c d^2 + 2m_w b^2 + I_c + 2I_m \ ,$$

where $\bar{\boldsymbol{M}}$, $\bar{\boldsymbol{V}}$ are the parameter matrices of the given CV system, $\boldsymbol{\tau} = [\tau_R \quad \tau_L]^T \in R^{2\times 1}$ is a control input vector, $m_t = m_c + 2m_w$ is a total mass of the CV, m_c is a mass of the body without driving wheels, m_w is a mass of each wheel, I_m is a moment of inertia of each motor, I_w is a moment of inertia of each wheel, and I_c is a moment of inertia of the body.

The CV model can be described by the following state function as:

$$\begin{cases} \dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \\ \boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} \end{cases}, \quad (6)$$

$$\boldsymbol{A} = -(\bar{\boldsymbol{M}}\boldsymbol{T}^{-1})^{-1}\bar{\boldsymbol{V}}\boldsymbol{T}^{-1}, \boldsymbol{B} = (\bar{\boldsymbol{M}}\boldsymbol{T}^{-1})^{-1}, \boldsymbol{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

where ***A, B,*** and ***C*** are parameter matrices of the CV,

$\boldsymbol{x} = \begin{bmatrix} V_A & \omega_A \end{bmatrix}^T$ is a system state vector, $\boldsymbol{y} = \begin{bmatrix} V_A & \omega_A \end{bmatrix}^T$ is an output vector, and $\boldsymbol{u} = \begin{bmatrix} \tau_R & \tau_L \end{bmatrix}^T \in R^{2\times1}$ is a torque input vector applied to the two wheels of the CV.

3. TRACKING CONTROLLER DESIGN

A global posture tracking error vector with respect to the global coordinate frame *OXY* is defined as follows:

$$\boldsymbol{e}_g = \boldsymbol{x}_r - \boldsymbol{x}_v \tag{7}$$

$$\boldsymbol{e}_g = \begin{bmatrix} e_X \\ e_Y \\ e_\theta \end{bmatrix} = \begin{bmatrix} X_r \\ Y_r \\ \theta_r \end{bmatrix} - \begin{bmatrix} X_A \\ Y_A \\ \theta_A \end{bmatrix} = \begin{bmatrix} X_r - X_A \\ Y_r - Y_A \\ \theta_r - \theta_A \end{bmatrix} \tag{8}$$

where $\boldsymbol{x}_v = \begin{bmatrix} X_A & Y_A & \theta_A \end{bmatrix}^T$ is a posture vector of the CV and $\boldsymbol{x}_r = \begin{bmatrix} X_r & Y_r & \theta_r \end{bmatrix}^T$ is a reference input vector with reference linear velocity V_r and angular velocity ω_r.

A local posture tracking error vector with respect to the local coordinate frame *Cxy* is defined as follows:

$$\boldsymbol{e} = \boldsymbol{R}\boldsymbol{e}_g = \boldsymbol{R}(\boldsymbol{x}_r - \boldsymbol{x}_v) \tag{9}$$

$$\boldsymbol{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} \cos\theta_A & \sin\theta_A & 0 \\ -\sin\theta_A & \cos\theta_A & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_r - X_A \\ Y_r - Y_A \\ \theta_r - \theta_A \end{bmatrix} \tag{10}$$

where $\boldsymbol{R}$ is the orthogonal rotational matrix.

By using Eqs. (1) and (2), the first-order time derivative of Eq. (10) can be obtained:

$$\dot{\boldsymbol{e}} = \begin{cases} \dot{e}_1 = V_r \cos e_3 - V_A + e_2\omega_A \\ \dot{e}_2 = V_r \sin e_3 - e_1\omega_A \\ \dot{e}_3 = \omega_r - \omega_A \end{cases} \tag{11}$$

A Lyapunov function candidate is chosen to analyze stability and determine control laws as follows:

$$V_0 = \frac{1}{2}e_1^2 + \frac{1}{2}e_2^2 + \frac{1}{k_2}(1 - \cos e_3) > 0 \tag{12}$$

where k_2 is a positive constant.

By using Eq. (11), the derivative of V_0 can be obtained as follows:

$$\begin{aligned} \dot{V}_0 &= e_1(-V_A + V_r \cos e_3) \\ &\quad + \frac{1}{k_2}(\sin e_3)(\omega_r - \omega_A + k_2 e_2 V_r) \end{aligned} \tag{13}$$

To satisfy the condition $\dot{V}_0 \le 0$, a virtual control input vector $\boldsymbol{\eta}$ is chosen as follows:

$$\boldsymbol{\eta} = \begin{bmatrix} V_A \\ \omega_A \end{bmatrix} = \begin{bmatrix} V_r \cos e_3 + k_1 e_1 \\ \omega_r + k_2 V_r e_2 + k_3 \sin e_3 \end{bmatrix} \tag{14}$$

where k_1, k_2, and k_3 are positive constants.

By substituting Eq. (14) into Eq. (13), the derivative of V_0 can be obtained as follows:

$$\dot{V}_0 = -k_1 e_2^2 - \frac{k_3}{k_2}(\sin e_3)^2 \le 0 \tag{15}$$

According to Lyapunov stability theory, if $V_0 > 0$ and $\dot{V}_0 \le 0$, the system is stable. And by Eqs. (10)~(15) and Barbalat's Lemma, the local posture tracking error vector $\boldsymbol{e}$ converges to zero as $t \to \infty$.

Since $\bar{\boldsymbol{M}}$ and $\bar{\boldsymbol{V}}$ of the given CV system in Eq. (5) are not measurable. Thus, $\boldsymbol{A}$, $\boldsymbol{B}$ matrices in Eq. (6) are uncertain parameters. A MRAC method is utilized to estimate these uncertain parameters. Otherwise, the reference input vector $\boldsymbol{\eta}$ of the MRAC is changed due to reference trajectory input changes. Hence, a reference model of the dynamic system model Eq. (6) is modified to its response tracking the reference velocity inputs accurately. Therefore, the modified reference model is given as follows:

$$\dot{\boldsymbol{x}}_m = \boldsymbol{A}_m \boldsymbol{x}_m + \boldsymbol{B}_m \boldsymbol{\eta} + \dot{\boldsymbol{\eta}} \tag{16}$$

where $\boldsymbol{A}_m \in R^{2\times2}$ is a known Hurwitz system matrix of the reference model, $\boldsymbol{B}_m \in R^{2x2}$ is a known input matrix of full rank of the reference model, $\boldsymbol{x}_m = \begin{bmatrix} V_{Am} & \omega_{Am} \end{bmatrix}^T$ is a reference model state vector, and $\boldsymbol{\eta}$ is considered as a reference velocity input vector of the reference model. There exists an unknown constant control gain matrix $\boldsymbol{K}_x \in R^{2\times2}$ and an unknown positive definite constant matrix $\boldsymbol{\Gamma} \in R^{2\times2}$ such that the following equations are satisfied:

$$\begin{cases} \boldsymbol{A} = \boldsymbol{A}_m - \boldsymbol{B}\boldsymbol{K}_x \\ \boldsymbol{B} = \boldsymbol{B}_m \boldsymbol{\Gamma} \end{cases} \tag{17}$$

A model error vector is defined as:

$$\boldsymbol{e}_m = \boldsymbol{x} - \boldsymbol{x}_m \tag{18}$$

By using Eq. (6) and Eqs. (16)~(18), the first time derivative of $\boldsymbol{e}_m$ is given as:

$$\begin{aligned}\dot{\boldsymbol{e}}_m &= \dot{\boldsymbol{x}} - \dot{\boldsymbol{x}}_m \\ &= \boldsymbol{A}_m\boldsymbol{e}_m + \boldsymbol{B}_m\boldsymbol{\Gamma}(\boldsymbol{u} - \boldsymbol{K}_x\boldsymbol{x} - \boldsymbol{K}_\eta\boldsymbol{\eta} - \boldsymbol{K}_h\dot{\boldsymbol{\eta}})\end{aligned} \tag{19}$$

where $\boldsymbol{K}_\eta = \boldsymbol{\Gamma}^{-1}$ and $\boldsymbol{K}_h = (\boldsymbol{B}_m\boldsymbol{\Gamma})^{-1}$ are unknown positive definite constant matrices.

Control input vector for stabilizing the system of Eq. (6) is designed:

$$\boldsymbol{u} = \hat{\boldsymbol{K}}_x\boldsymbol{x} + \hat{\boldsymbol{K}}_\eta\boldsymbol{\eta} + \hat{\boldsymbol{K}}_h\dot{\boldsymbol{\eta}} \tag{20}$$

where $\hat{\boldsymbol{K}}_x, \hat{\boldsymbol{K}}_\eta, and\ \hat{\boldsymbol{K}}_h$ are estimation parameter matrices of $\boldsymbol{K}_x, \boldsymbol{K}_\eta, and\ \boldsymbol{K}_h$.

By using the control input vector of Eq. (20), the first time derivative of Eq. (13) becomes:

$$\begin{aligned}\dot{\boldsymbol{e}}_m &= \boldsymbol{A}_m\boldsymbol{e}_m + \boldsymbol{B}_m\boldsymbol{\Gamma}(\hat{\boldsymbol{K}}_x\boldsymbol{x} + \hat{\boldsymbol{K}}_\eta\boldsymbol{\eta} + \hat{\boldsymbol{K}}_h\dot{\boldsymbol{\eta}} - \boldsymbol{K}_x\boldsymbol{x} - \boldsymbol{K}_\eta\boldsymbol{\eta} - \boldsymbol{K}_h\dot{\boldsymbol{\eta}}) \\ &= \boldsymbol{A}_m\boldsymbol{e}_m + \boldsymbol{B}_m\boldsymbol{\Gamma}[(\hat{\boldsymbol{K}}_x - \boldsymbol{K}_x)\boldsymbol{x} + (\hat{\boldsymbol{K}}_\eta - \boldsymbol{K}_\eta)\boldsymbol{\eta} + (\hat{\boldsymbol{K}}_h - \boldsymbol{K}_h)\dot{\boldsymbol{\eta}}] \\ &= \boldsymbol{A}_m\boldsymbol{e}_m + \boldsymbol{B}_m\boldsymbol{\Gamma}(\tilde{\boldsymbol{K}}_x\boldsymbol{x} + \tilde{\boldsymbol{K}}_\eta\boldsymbol{\eta} + \tilde{\boldsymbol{K}}_h\dot{\boldsymbol{\eta}})\end{aligned} \tag{21}$$

where

$\tilde{\boldsymbol{K}}_x = \hat{\boldsymbol{K}}_x - \boldsymbol{K}_x$, $\tilde{\boldsymbol{K}}_\eta = \hat{\boldsymbol{K}}_\eta - \boldsymbol{K}_\eta$, $\tilde{\boldsymbol{K}}_h = \hat{\boldsymbol{K}}_h - \boldsymbol{K}_h$.

Theorem 1. A MRAC system defined by Eqs. (6) and (16) is stable as long as the control input vector of Eq. (20) and update laws are given as:

$$\begin{cases}\dot{\tilde{\boldsymbol{K}}}_x = \dot{\hat{\boldsymbol{K}}}_x = -\boldsymbol{\gamma}_1\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{x}^T \\ \dot{\tilde{\boldsymbol{K}}}_\eta = \dot{\hat{\boldsymbol{K}}}_\eta = -\boldsymbol{\gamma}_2\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{\eta}^T \\ \dot{\tilde{\boldsymbol{K}}}_h = \dot{\hat{\boldsymbol{K}}}_h = -\boldsymbol{\gamma}_3\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\dot{\boldsymbol{\eta}}^T\end{cases} \tag{22}$$

where $\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2$, and $\boldsymbol{\gamma}_3$ are adaptation gain matrices, a positive definite matrix $\boldsymbol{P}$ is a solution of Lyapunov equation as follows:

$$\boldsymbol{A}_m^T\boldsymbol{P} + \boldsymbol{P}\boldsymbol{A}_m = -\boldsymbol{Q} \tag{23}$$

where $\mathbf{P} = diag\left(\begin{bmatrix} p_{11} & p_{22}\end{bmatrix}\right)$, $\mathbf{Q} = diag\left(\begin{bmatrix} q_{11} & q_{22}\end{bmatrix}\right)$ with $p_{11}, p_{22}, q_{11}, q_{22} > 0$.

Proof of Theorem 1. A Lyapunov function candidate is chosen to analyze the stability of the system in Eq. (6) as follows:

$$\begin{aligned}V &= \frac{1}{2}\boldsymbol{e}_m^T\boldsymbol{P}\boldsymbol{e}_m + \frac{1}{2}trace\left(\tilde{\boldsymbol{K}}_x^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_1^{-1}\tilde{\boldsymbol{K}}_x\right) \\ &+ \frac{1}{2}trace\left(\tilde{\boldsymbol{K}}_\eta^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_2^{-1}\tilde{\boldsymbol{K}}_\eta\right) \\ &+ \frac{1}{2}trace\left(\tilde{\boldsymbol{K}}_h^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_3^{-1}\tilde{\boldsymbol{K}}_h\right) \geq 0\end{aligned} \tag{24}$$

By using Eqs. (21) and (23), the first time derivative of V in Eq. (24) is given as:

$$\begin{aligned}\dot{V} &= \frac{1}{2}(\dot{\boldsymbol{e}}_m^T\boldsymbol{P}\boldsymbol{e}_m + \boldsymbol{e}_m^T\boldsymbol{P}\dot{\boldsymbol{e}}_m) + trace\left(\tilde{\boldsymbol{K}}_x^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_1^{-1}\dot{\tilde{\boldsymbol{K}}}_x\right) \\ &+ trace\left(\tilde{\boldsymbol{K}}_\eta^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_2^{-1}\dot{\tilde{\boldsymbol{K}}}_\eta\right) + trace\left(\tilde{\boldsymbol{K}}_h^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_3^{-1}\dot{\tilde{\boldsymbol{K}}}_h\right) \\ &= -\frac{1}{2}\boldsymbol{e}_m^T\boldsymbol{Q}\boldsymbol{e}_m + trace(\tilde{\boldsymbol{K}}_x^T\boldsymbol{\Gamma}\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{x}^T) \\ &+ trace(\tilde{\boldsymbol{K}}_\eta^T\boldsymbol{\Gamma}\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{\eta}^T) + trace(\tilde{\boldsymbol{K}}_h^T\boldsymbol{\Gamma}\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\dot{\boldsymbol{\eta}}^T) \\ &+ trace\left(\tilde{\boldsymbol{K}}_x^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_1^{-1}\dot{\tilde{\boldsymbol{K}}}_x\right) + trace\left(\tilde{\boldsymbol{K}}_\eta^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_2^{-1}\dot{\tilde{\boldsymbol{K}}}_\eta\right) \\ &+ trace\left(\tilde{\boldsymbol{K}}_h^T\boldsymbol{\Gamma}\boldsymbol{\gamma}_3^{-1}\dot{\tilde{\boldsymbol{K}}}_h\right) \\ &= -\frac{1}{2}\boldsymbol{e}_m^T\boldsymbol{Q}\boldsymbol{e}_m + trace[\tilde{\boldsymbol{K}}_x^T\boldsymbol{\Gamma}(\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{x}^T + \boldsymbol{\gamma}_1^{-1}\dot{\tilde{\boldsymbol{K}}}_x)] \\ &+ trace[\tilde{\boldsymbol{K}}_\eta^T\boldsymbol{\Gamma}(\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\boldsymbol{\eta}^T + \boldsymbol{\gamma}_2^{-1}\dot{\tilde{\boldsymbol{K}}}_\eta)] \\ &+ trace[\tilde{\boldsymbol{K}}_h^T\boldsymbol{\Gamma}(\boldsymbol{B}_m^T\boldsymbol{P}\boldsymbol{e}_m\dot{\boldsymbol{\eta}}^T + \boldsymbol{\gamma}_3^{-1}\dot{\tilde{\boldsymbol{K}}}_h)]\end{aligned} \tag{25}$$

By using the update laws of Eqs. (22), the first time derivative of V of Eq. (25) becomes:

$$\dot{V} = -\frac{1}{2}\boldsymbol{e}_m^T\boldsymbol{Q}\boldsymbol{e}_m \leq 0 \tag{26}$$

From Eqs. (24) and (26), $\hat{\boldsymbol{K}}_x, \hat{\boldsymbol{K}}_\eta, and\ \hat{\boldsymbol{K}}_h$ are bounded. Using Eqs. (22)~(26) and since elements of $\boldsymbol{A}_m$ and $\boldsymbol{B}_m$ are chosen as conditions of Eq. (17), based on Barbalat's Lemma, $\lim_{t\to\infty}\boldsymbol{e}_m \to 0\ as\ t \to \infty$. Thus, from Eq. (18), it means $\boldsymbol{x} \to \boldsymbol{x}_m\ \ as\ t \to \infty$.

The configuration of the tracking controller can be described as shown in Fig. 2.

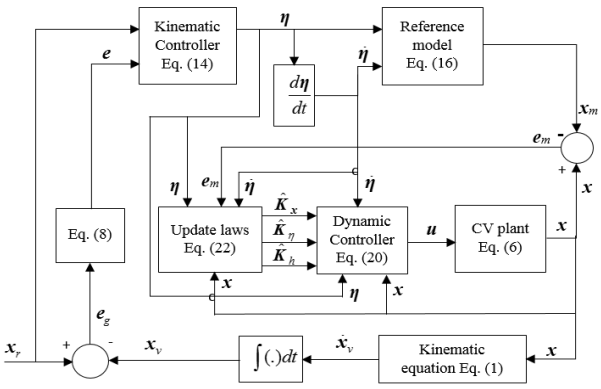
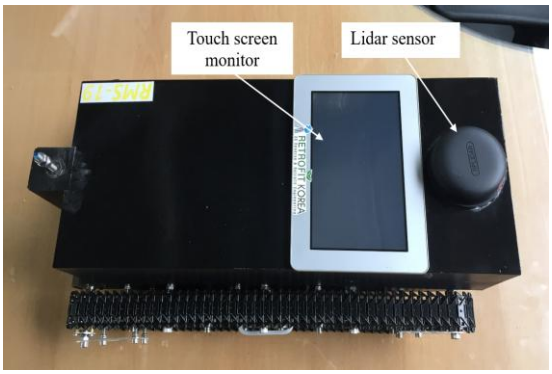


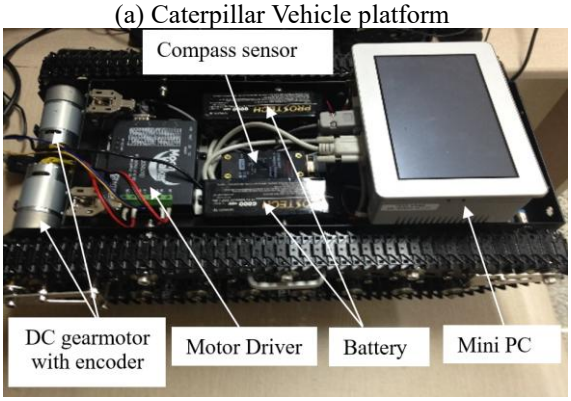
Fig. 2 Configuration of the MRAC tracking controller

4. SIMULATION AND EXPERIMENTAL RESULTS

To verify the effectiveness of the proposed MRAC controller, the simulations and experiments have been done for the CV to track the desired trajectory.



(a) Caterpillar Vehicle platform



(b) Internal structure of the CV

Fig. 3 Configuration of the CV

Configuration of the CV uses in this paper is shown as Fig. 3. These experimental results of the proposed MRAC controller are compared with simulation results.

4.1 Parameters of the controller

Tables 1-3 show the parameters of the CV, the initial values and parameter values of the proposed tracking controller.

Table 1 Parameters values of the CV

Parameters	Descriptions	Values
$2b$	Distance of wheels	0.24m
r	Radius of wheel	0.04m
m_c	Mass of CV	20kg

Table 2 Parameter values of the proposed controller

Paras.	Values
$\boldsymbol{A}_m, \boldsymbol{B}_m$	$\boldsymbol{A}_m = \begin{bmatrix} -50 & 0 \\ 0 & -50 \end{bmatrix}, \boldsymbol{B}_m = \begin{bmatrix} 50 & 0 \\ 0 & 50 \end{bmatrix}$
γ_1, γ_2 γ_3	$\gamma_1 = \gamma_2 = \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}, \gamma_3 = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$
$\boldsymbol{P}$	$\begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}$
V_r	0.1m/s
ω_r	0.1256rad/s
k_1	$50s^{-1}$
k_2	$100rad/m^2$
k_3	40rad/s

Table 3 Initial values of the proposed controller

Parameters	Values
$\boldsymbol{x}_v(0)$	$[0 \quad 0 \quad 0]^T$
$\boldsymbol{x}(0)$	$[0 \quad 0]^T$

4.2 Simulation and experimental results

In this experiment, the posture of CV (X_A, Y_A, θ_A) with respect to the global coordinate frame is obtained by the SLAM algorithm in [5]. Fig. 4 shows the desired global trajectory reference with a black continuous line, the global trajectory tracking result by simulation with a magenta dashed line, and the global trajectory tracking result by the experiment with a blue center line.

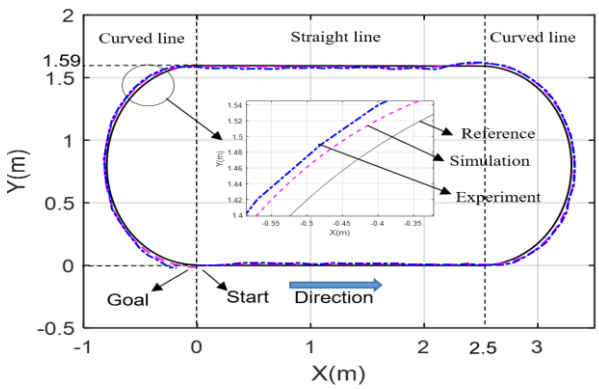


Fig. 4 Global trajectory tracking results of the CV

The local posture tracking error vector $\boldsymbol{e}$ of the CV with respect to the local coordinate frame using the proposed controller in the experiment and the simulation are shown in Figs. 5 - 7. Fig. 5 shows the local tracking errors e_1 of the CV on x axis. The local tracking error e_1 of the CV on x axis in the experiment converges to zero after 5 seconds and it is bounded from −0.006m to +0.009m, while the local tracking error e_1 of the CV on x axis in the simulation converges to zero after 5 seconds and it is bounded from 0m to +0.003m.

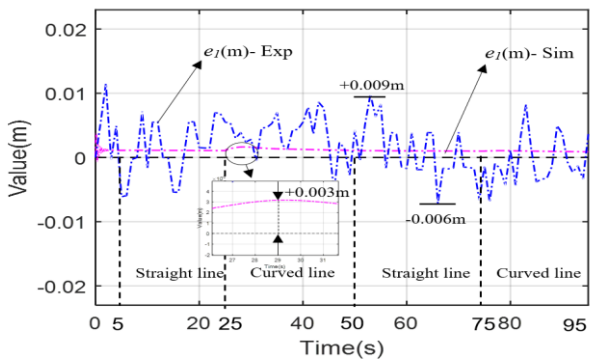


Fig. 5 Loal tracking error e_1 of the CV on *x axis*

Fig. 6 shows local tracking errors e_2 of the CV on y axis. The local tracking error e_2 of the CV on y axis in the experiment converges to zero after 25 seconds and it is bounded from −0.015m to +0.024m, while the local tracking error e_2 of the CV on y axis in the simulation converges to zero after 25 seconds and it is bounded from −0.006m to +0.016m.

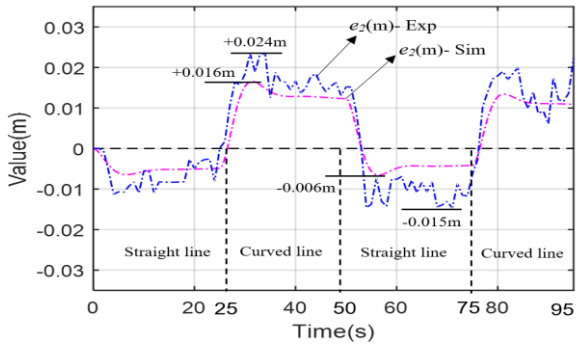


Fig. 6 Local tracking error e_2 of the CV on y axis

Fig. 7 shows the local orientation tracking error e_3 of the CV. The local orientation tracking error e_3 of the CV in the experiment converges to zero after 8 seconds and it is bounded from −0.065rad to +0.06rad around zero, while the local orientation tracking error e_3 of the CV in the simulation converges to zero and after 5 seconds it is bounded from −0.01rad to +0.01rad. However, the local orientation tracking error e_3 of the CV in the both simulation and experiment has big errors at t_1 = 25s, t_2 = 50s, and t_3 = 75s due to the reference changes.

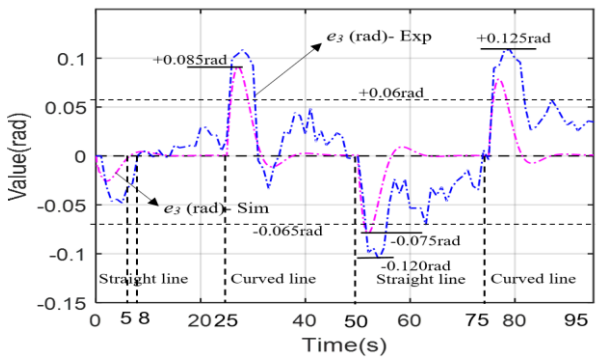


Fig. 7 Local orientation tracking error e_3 of the CV

The global posture tracking error vector $\boldsymbol{e}_g$ of the CV with respect to the global coordinate frame in the experiment and the simulation are shown in Figs. 8 - 10. Fig. 8 shows the global tracking error e_X of the CV on X axis. The global tracking error e_X of the CV on X axis in the experiment converges to zero after 5 seconds and it is bounded from −0.018m to +0.016m around zero, while the global tracking error e_X of the CV on X axis in the simulation converges to zero after 5 seconds and it is bounded from −0.014m to +0.012m around zero. Fig. 9 shows the global tracking error e_Y of the CV on Y axis. The global tracking error e_Y of the CV on Y axis in the experiment converges to zero after 25 seconds and it is bounded from −0.017m to +0.018m around zero, while the global tracking error e_Y of the CV on Y axis in the simulation converges to zero after 25 seconds and it is bounded from −0.012m to +0.014m around zero. These figures show the global posture tracking error results of the CV in the straight line are better than the that in curved line. Fig. 10 shows the global orientation tracking error e_θ of the CV. The global orientation tracking error e_θ of the CV in the experiment converges to zero after 8 seconds and it is bounded from −0.065rad to +0.06rad around zero, while the global orientation tracking error e_θ of the CV in the simulation converges to zero after 5 seconds and it is bounded from −0.01rad to +0.01rad around zero.

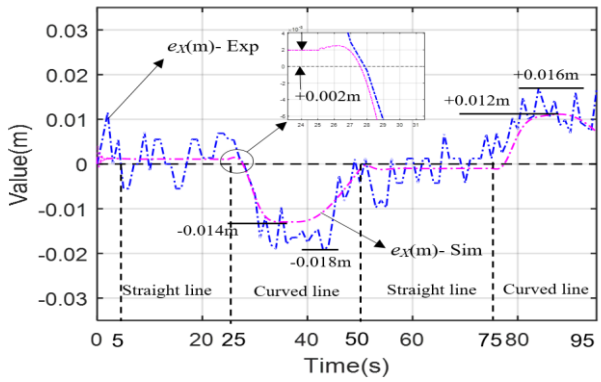


Fig. 8 Global tracking error e_X of the CV on X *axis*

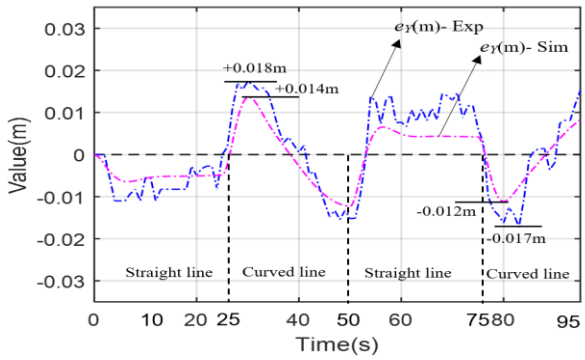


Fig. 9 Global tracking error e_Y of the CV on Y *axis*

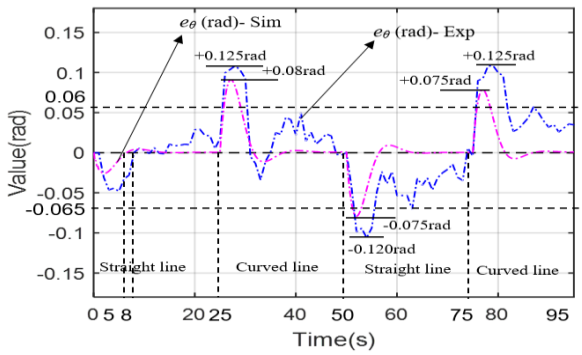


Fig. 10 Global orientation tracking error e_θ of the CV

Fig. 11 shows the model error vector $\boldsymbol{e}_m$ of the MRAC. The model error e_{m1} of the MRAC in experiment is bounded in the range from −0.007m/s to +0.006m/s, while the model error e_{m1} of the MRAC in simulation converges to zero. The model error e_{m2} of the MRAC in experiment is bounded in the range from −0.014rad/s to +0.007rad/s, while the model error e_{m2} of the MRAC in simulation is bounded in the range from −0.004rad/s to +0.002rad/s. Therefore, it is shown that the given reference model matching with the real dynamic model well.

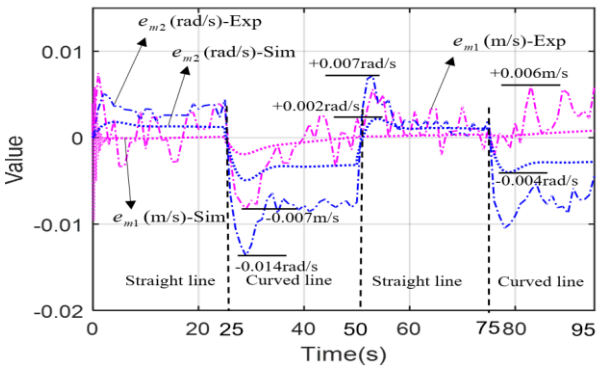


Fig. 11 Model error vector $\boldsymbol{e}_m$

Fig. 12 shows the control input vector $\boldsymbol{u}$ of the MRAC. The control input vector u_1 of the MRAC in experiment is bounded in the range from +0.15Nm to +0.28Nm, while the control input vector u_1 of the MRAC in simulation converges to +0.14Nm. The control input vector u_2 of the MRAC in experiment is bounded in the range from −0.29Nm to +0.30Nm, while the control input vector u_2 of the MRAC in simulation is bounded in the range from −0.18Nm to +0.19Nm.

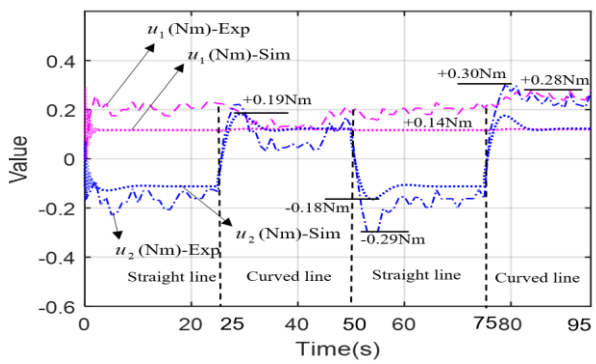


Fig. 12 Control input vector $\boldsymbol{u}$

Figs. 13-15 show the update values of the proposed controller $\hat{K}_x$, $\hat{K}_\eta$, and $\hat{K}_h$ in experiment, respectively. Fig. 13 shows the update values $\hat{K}_{x11}$, $\hat{K}_{x12}$, $\hat{K}_{x21}$, and $\hat{K}_{x22}$. The update value $\hat{K}_{x11}$ is bounded in the range from −0.010Vs/m to +0.015Vs/m. The update value $\hat{K}_{x12}$ is bounded in the range from −0.005 Vs/rad to +0.020Vs/rad. The update value $\hat{K}_{x21}$ is bounded in the range from −0.010Vs/rad to +0.005Vs/rad around −0.1Vs/rad. The update value $\hat{K}_{x22}$ is bounded in the range from −0.005Vs/m to +0.015Vs/m around −0.1Vs/m. Fig. 14 shows the update values $\hat{K}_{\eta11}$, $\hat{K}_{\eta12}$, $\hat{K}_{\eta21}$, and $\hat{K}_{\eta22}$. The update value $\hat{K}_{\eta11}$ is bounded in the range from −0.013Vs/m to +0.015Vs/m around 0.1Vs/m. The update value $\hat{K}_{\eta12}$ is bounded in the range from −0.016Vs/rad to 0.005Vs/rad around zero. The update value $\hat{K}_{\eta21}$ is bounded in the range from −0.020Vs/rad to 0.008Vs/rad around 0.1Vs/rad. The update value $\hat{K}_{\eta22}$ is bounded in the range from −0.005Vs/m to +0.021Vs/m around 0.1Vs/m. Fig. 15 shows the update values $\hat{K}_{h11}$, $\hat{K}_{h12}$, $\hat{K}_{h21}$, and $\hat{K}_{h22}$. The update value $\hat{K}_{h11}$ is bounded in the range from -0.24×10^{-3} Vs²/m to $+0.3\times10^{-3}$ Vs²/m around 2×10^{-3} Vs²/m. The update value $\hat{K}_{h12}$ is bounded in the range from -3.5×10^{-4} Vs²/rad to $+8\times10^{-4}$ Vs²/rad around zero. The update value $\hat{K}_{h21}$ is bounded in the range from -0.4×10^{-3} Vs²/rad to $+0.17\times10^{-3}$ Vs²/rad around 2×10^{-3} Vs²/rad. The update value $\hat{K}_{h22}$ is bounded in the range from -0.1×10^{-3} Vs²/m to $+0.41\times10^{-3}$ Vs²/m around 2×10^{-3} Vs²/m.

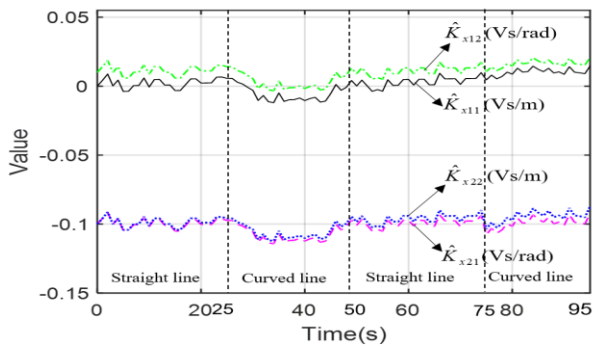


Fig. 13 Adaptable parameters of $\hat{K}_x$

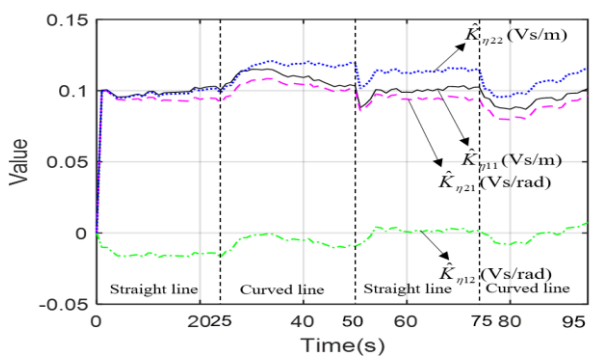


Fig. 14 Adaptable parameters of $\hat{K}_\eta$

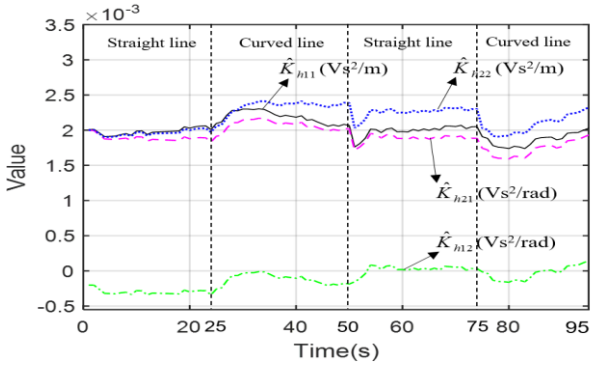


Fig. 15 Adaptable parameters of $\hat{K}_h$

5. CONCLUSION

A tracking controller design for Caterpillar Vehicle to track desired trajectory reference using backstepping-based MRAC was proposed. In order to verify the effectiveness of the proposed methods, the simulation and experimental results were presented. The simulation and experimental results showed the proposed method had adequate tracking performance for the desired trajectory reference. These experimental results were compared with those of the simulation results using the proposed method. The global tracking error e_X of the CV on X axis in the experiment converged to zero after 5 seconds and it was bounded from −0.018m to +0.016m around zero. The global tracking error e_Y of the CV on Y axis in the experiment converged to zero after 25 seconds and it was bounded from −0.017m to +0.018m around zero. The global orientation tracking error e_θ of the CV in the experiment converged to zero after 8 seconds and it was bounded from −0.065rad to +0.06rad around zero. The model error vector of the MRAC were bounded. It was shown that the given reference model matching with the real dynamic model well.

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